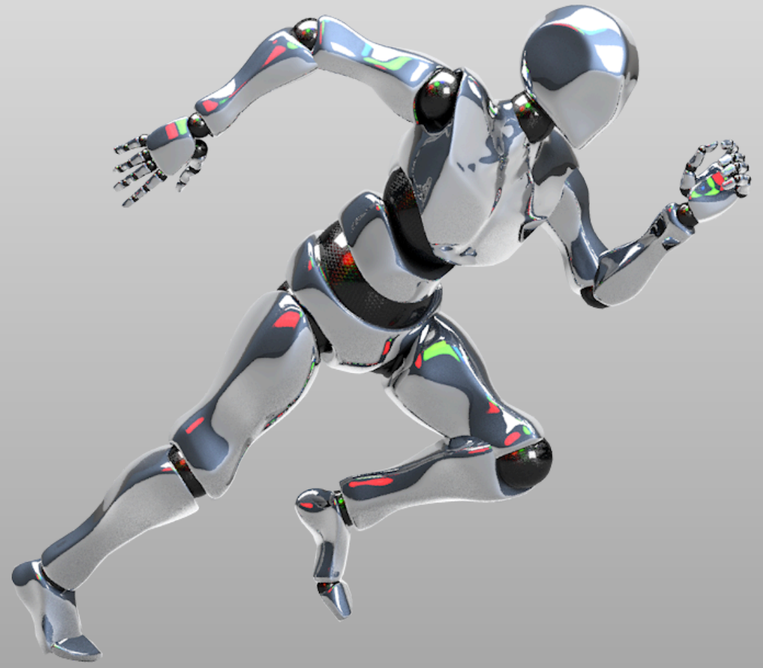
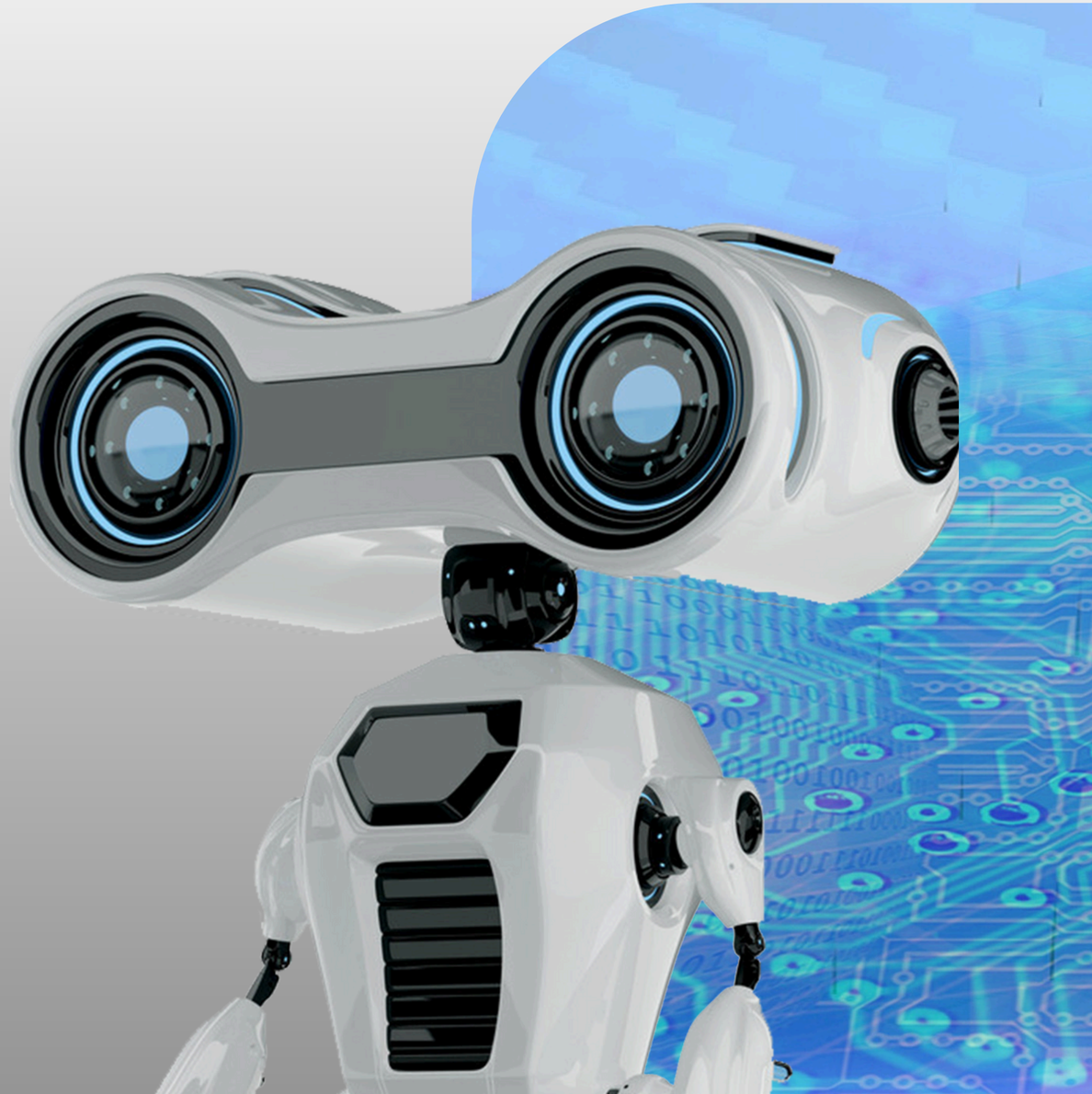
# Robot Operating System 2 : Architecture, Applications & Microcontroller Compatibility



Abdulmanan
Awais Khan
Muhammad Rohab

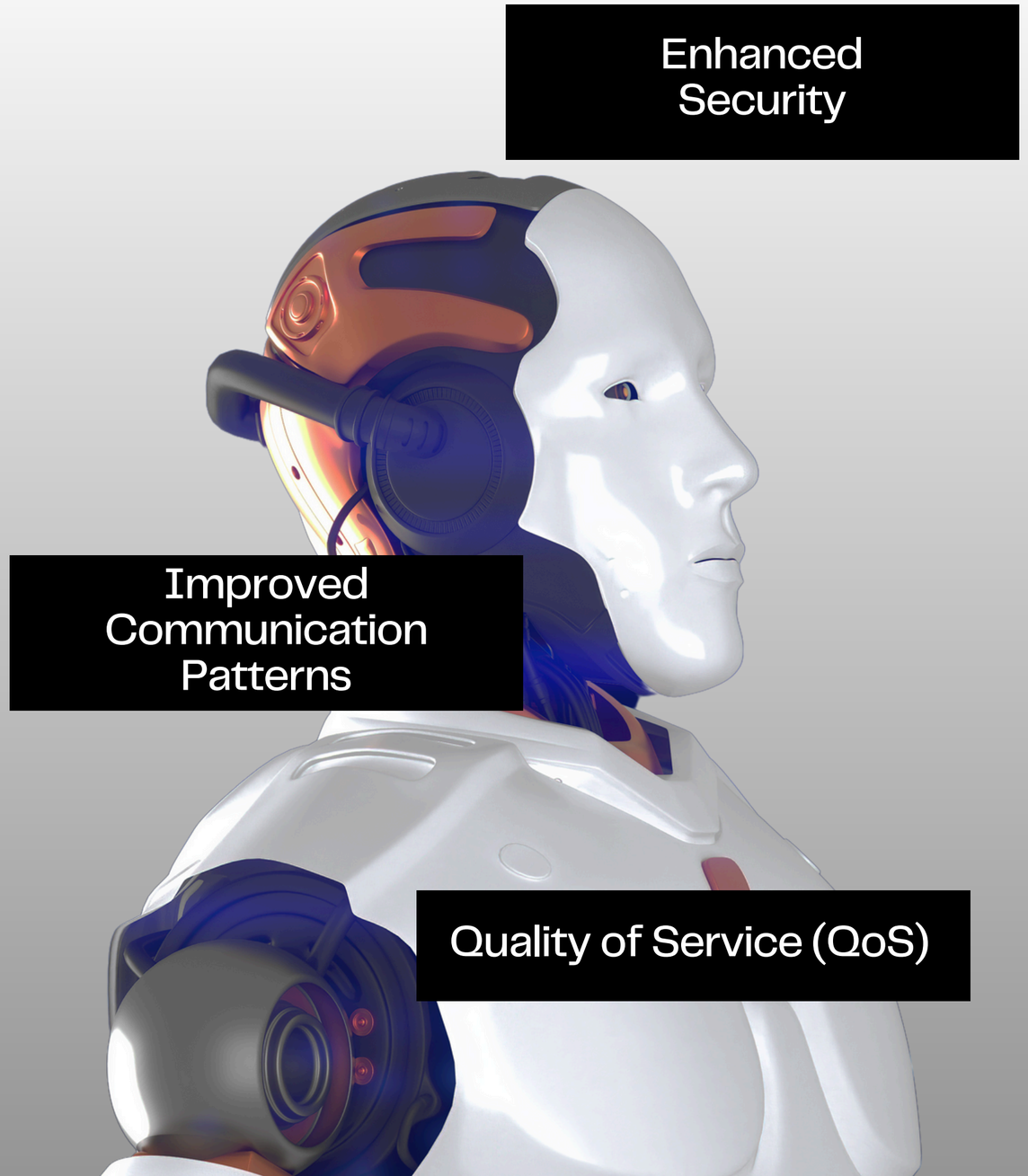
# Introduction & Motivation

- Robotics systems have become increasingly complex, requiring advanced software frameworks
- To tackle these challenges, ROS came, initially developed by two PhD students from Stanford University.
- ROS 1 became widely used but lacked determinism, security & industrial readiness.
- ROS 2 addresses these limitations using DDS, modular architecture & RTOS compatibility.



# ROS–I vs ROS–II

| | ROS–I | ROS–II |
|---|---|---|
| Architecture | single point of failure | robust, peer-to-peer communication |
| Security | No native security | authentication, encryption & access control |
| Embedded Systems | Not suitable for microcontrollers | Supports microcontrollers via micro–ROS |
| Multi–Platform & Multi–Robot | Constrained; weak multi–robot support | Multi–robot, multi–platform (Linux, Windows, macOS,) |

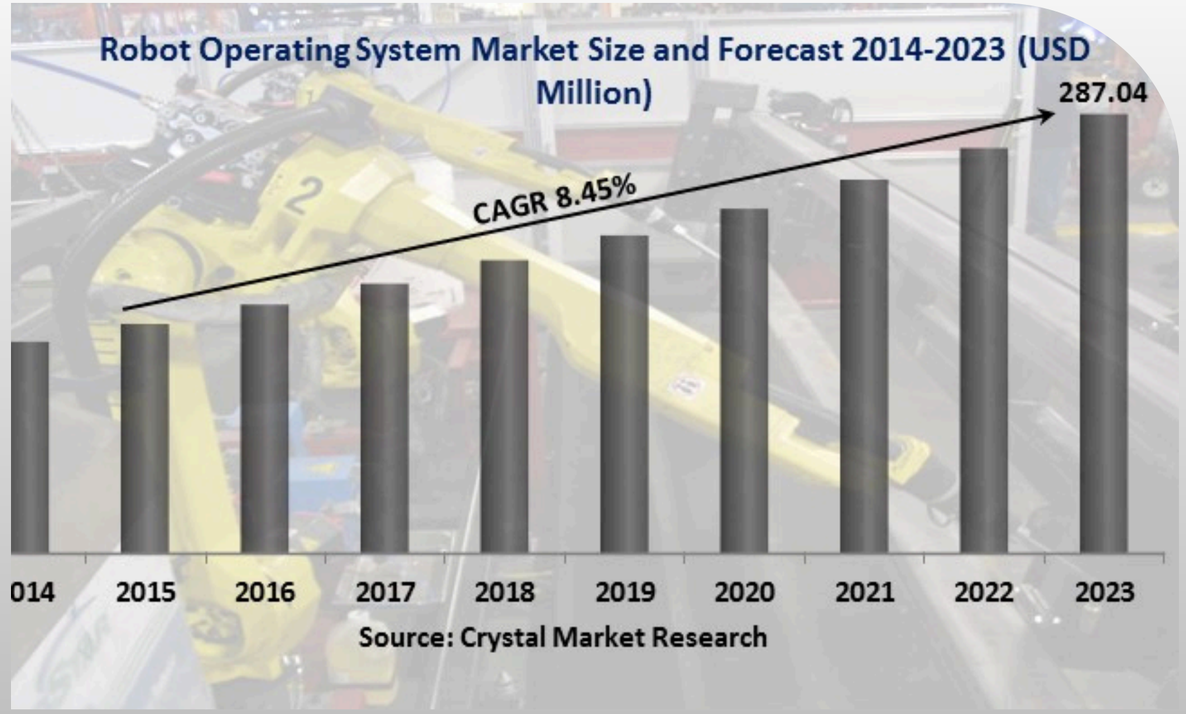



# ROS-II Architecture Overview

- **DDS Middleware**: Real-time, QoS profiles, peer-to-peer, no master node
- **Nodes, Topics, Services, Actions:** Communication patterns
- **Lifecycle Nodes:** Managed states for safety & predictability
- **Colcon Build System:** Modern multi-package build tool
- **Tools:** Gazebo, RViz2, rqt, Foxglove

# ROS 2 Applications

- **Autonomous Vehicles:** Sensor fusion, navigation, Autoware.Auto
- **Drones & UAVs:** Distributed control, micro-ROS integration
- **Industrial Automation:** AMRs, cobots, manipulators
- **Healthcare Robotics:** Assistive robots, surgical systems
- **Logistics & Warehousing:** Fleet management, AMRs



# Compatibility With Microcontrollers (micro-ROS)

- Full ROS 2 cannot run on MCUs → micro-ROS solves this.
- Runs on FreeRTOS, Zephyr, NuttX, Arduino framework.
- Architecture: MCU (client) ↔ micro-ROS Agent ↔ ROS 2 network

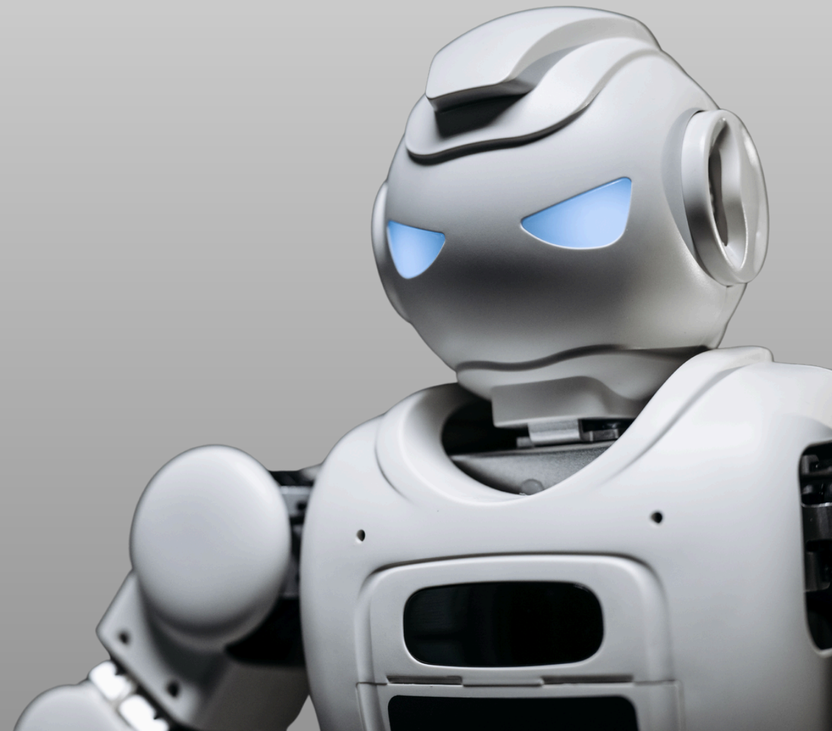
Supported MCUs:

- ESP32
- STM32 / Nucleo
- RP2040 (Raspberry Pi Pico)
- Arduino Portenta H7



# Conclusion

- ROS 2 is a major upgrade over ROS 1: real-time, secure, scalable.
- Widely adopted across research & industry.
- micro-ROS enables embedded/IoT devices to join ROS 2 systems.
- Future directions: micro-ROS expansion, AI integration, improved real-time execution.



# Thank You

